

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Nandhini sri V
Register Number	192521344

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I _NANDHINI SRI V(192521344)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

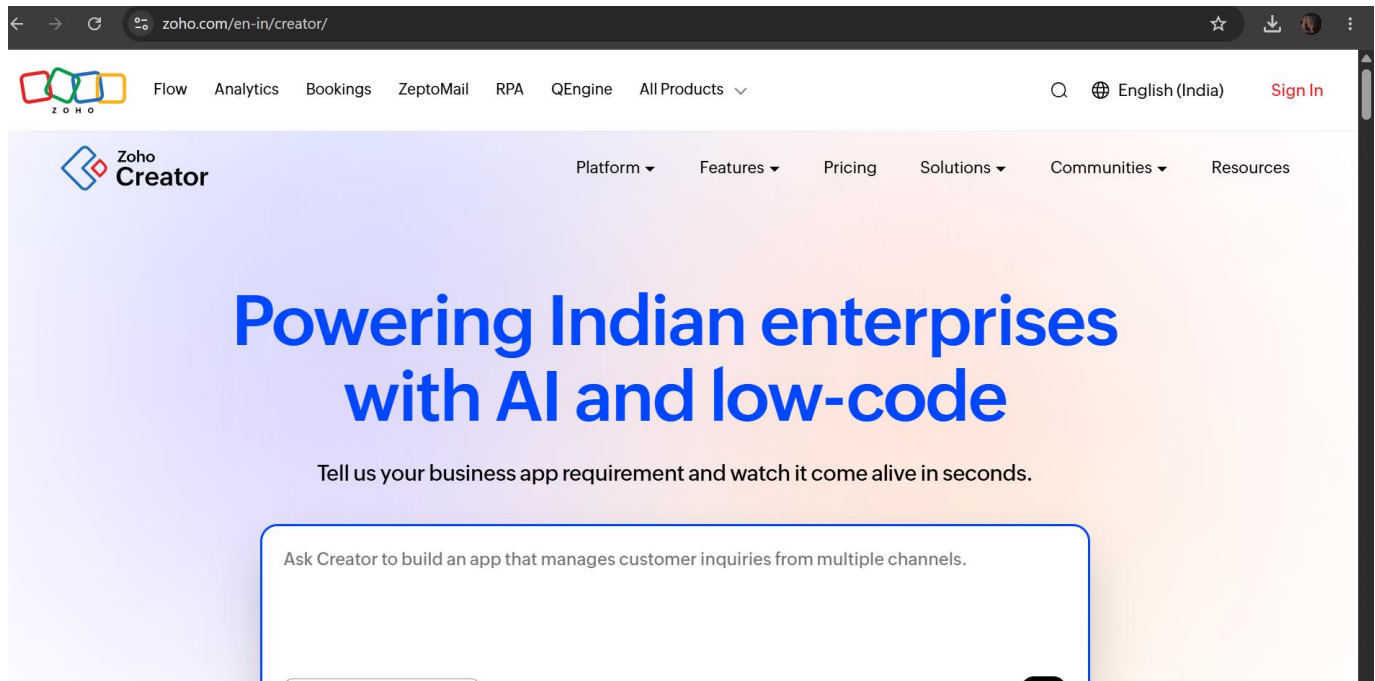
ANS:

AIM:

To create a simple cloud software application for Library book reservation system for SIMATS library and provide it as a service using any cloud service provider to demonstrate software as a service (saas).

PROCEDURE:

STEP1: Log into zoho.com



STEP 2: Create new application from scratch

Create from scratch

Application Name *

SIMATS Library Book Reservation System

☐ Enable Environment [Learn More](#)

Create Application

STEP 3: Create new application and create form from scratch

creator.zoho.in/appbuilder/nandy.nandhinisri22/simats-library-book-reservation-system/edit

SIMATS Library Book Reservation System

Design Workflow Settings Upgrade Access this application

SIMATS Library Book Reservation System

Start building your application by creating a form

Create New Form

How would you like to create your form?

On my own

From scratch

Import with data

Using Zia

From a template

Cab Booking

Conf Room Booking

Appointment Booking

Hotel Room Booking

Complaints

50 +
View More

Chats

Contacts

Here is your Smart Chat (Ctrl+Space)

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Get started faster!

Help

STEP 4: Create Bookings Form

SIMATS Library Book Reservation ...
SIMATS library book reser ...

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Name

Phone

Date

Email

Time

Zia

Field Properties

Field name

Name

Field link name

Name

View Field References

Validation

Mandatory

Display Fields

Prefix

First Name

Last Name

Chats

Contacts

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Help

STEP 5: Creating library Form

SIMATS Library Book Reservation...
SIMATS library book reser ...

+

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Name

Phone

Date

Email

Time

2/0

Field Properties

Field name

Name

Field link name

Name

View Field References

Validation

Mandatory

Display Fields

Prefix

First Name

Last Name

Chats

Connects

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Help

STEP 6: Creating Book Reservation form

SIMATS Library Book Reservation ...
book reservation form

+

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123
Number

15
Date

Name

Email

Address

Phone

123
Number

Date





Field Properties

Field name

Date

Field link name

Date_field

[View Field References](#)

Validation

☐ Mandatory ☐ No duplicate values

Initial value

dd-MMM-yyyy



Allowed Days

All days

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Help

STEP 7: Creating book inventory report

Library Management System
Feedback Collection

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

D

Student

Book

Rating

Comments

Feedback Date

Field Properties

Field name

Field link name

Validation

Initial value

Character maximum

Chats

Contacts

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Help

STEP 8: Create Feedback Form

Library Management System
Feedback Collection

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

D

Student

Book

Rating

Comments

Feedback Date

Field Properties

Field name

Field link name

Validation

Initial value

Character maximum

Chats

Contacts

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Help

STEP 10: Add sample data to all the created forms.

Library Management System

Book Details

Book Id: 1467

Title:

Author: Harry potter

Publication:

Year: #####

Edition:

Isbn:

Total Copies: #####

Available Copies: #####

Dashboard

Book Reservations

Book Transactions

Book Details

+ Book Details

Book Details Report

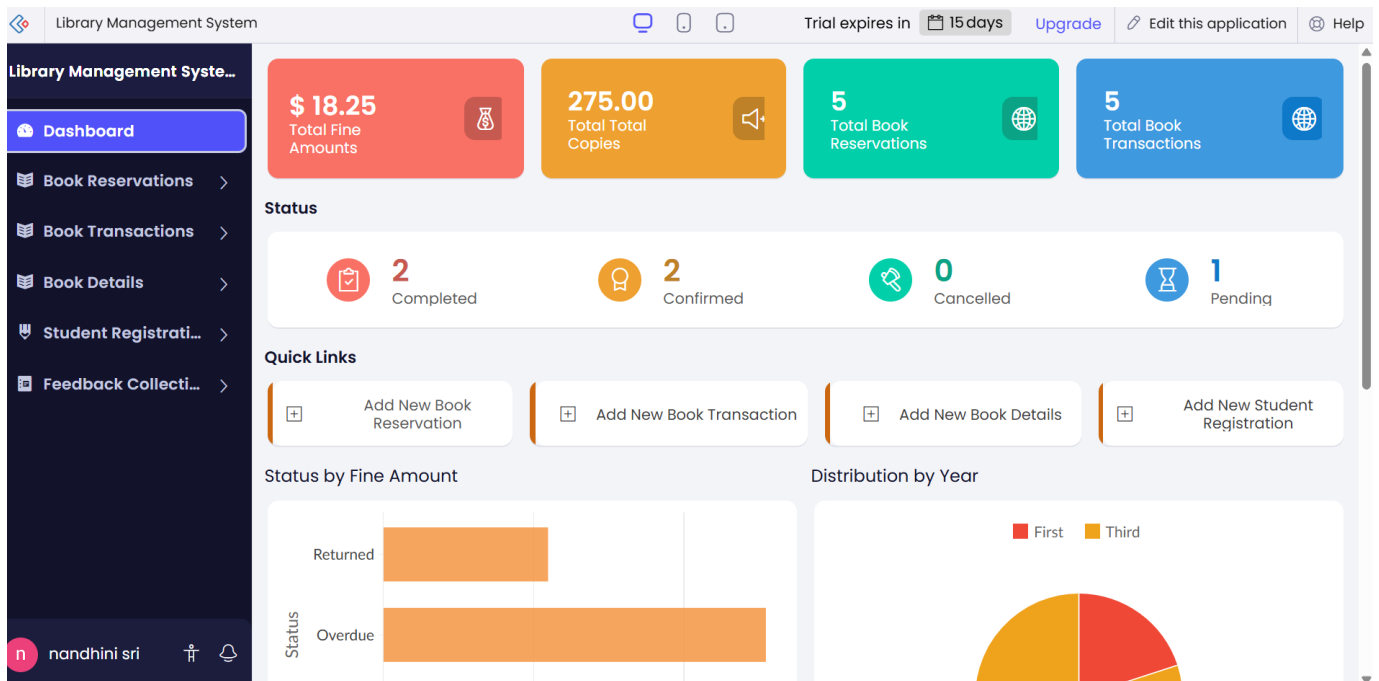
Categories

Student Registrati...

Feedback Collecti...

nandhini sri

STEP 11: Create and display the Dashboard with important book reservations, buying and student registration and feed back collections



Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

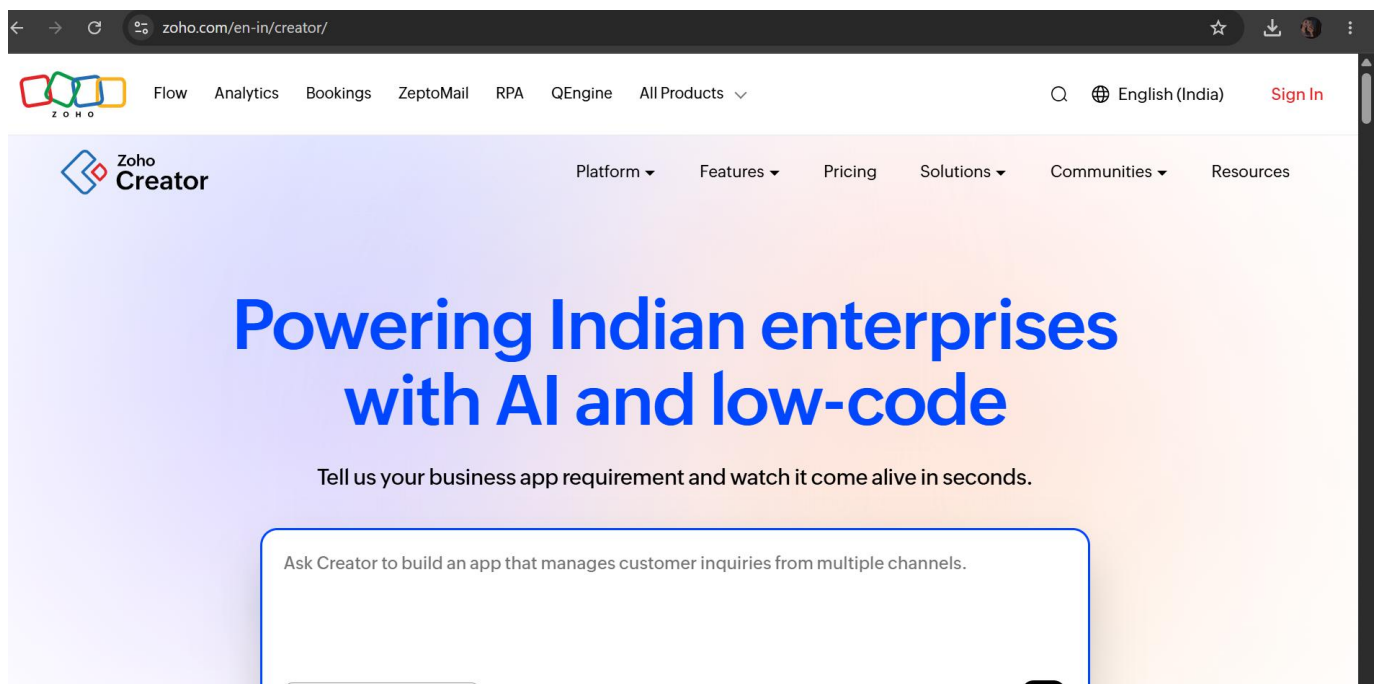
ANS:

AIM:

To create a simple cloud software application for Payroll processing system and provide it as a service using any cloud service provider to demonstrate software as a service (saas).

PROCEDURE:

STEP1: Log into zoho.com



STEP 2: Create new application from scratch

Create from scratch

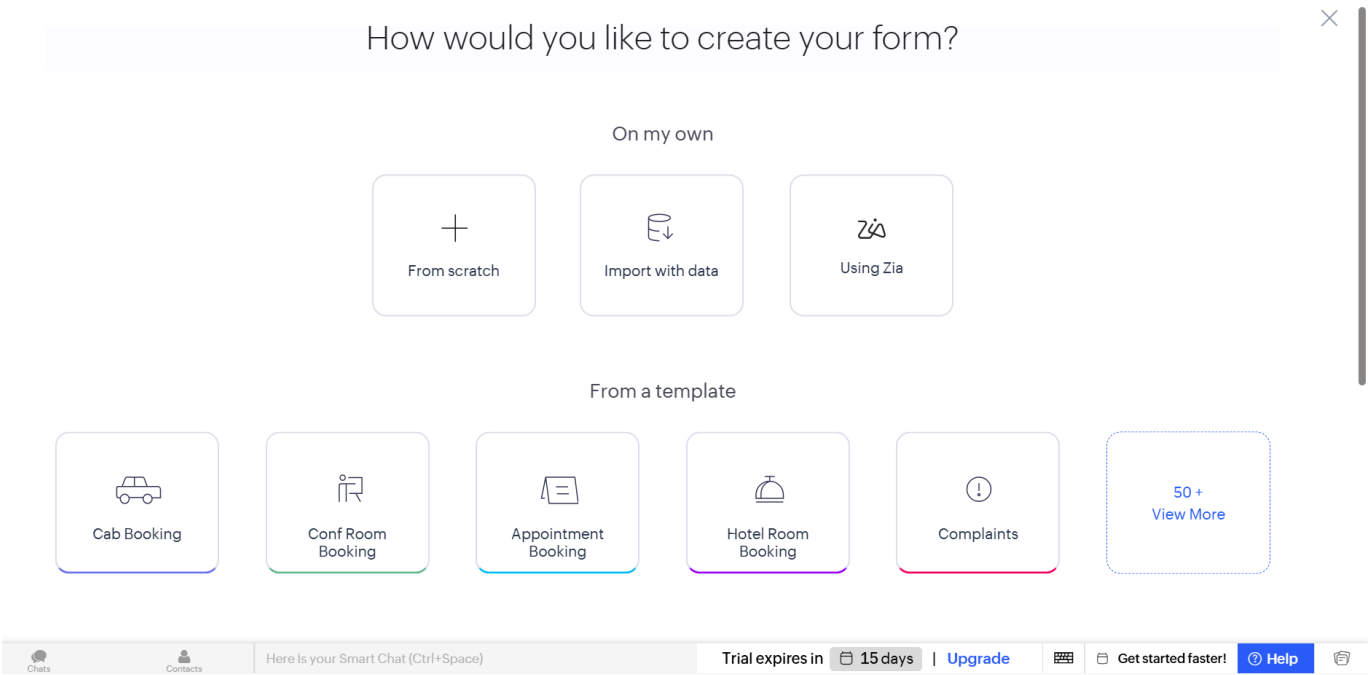
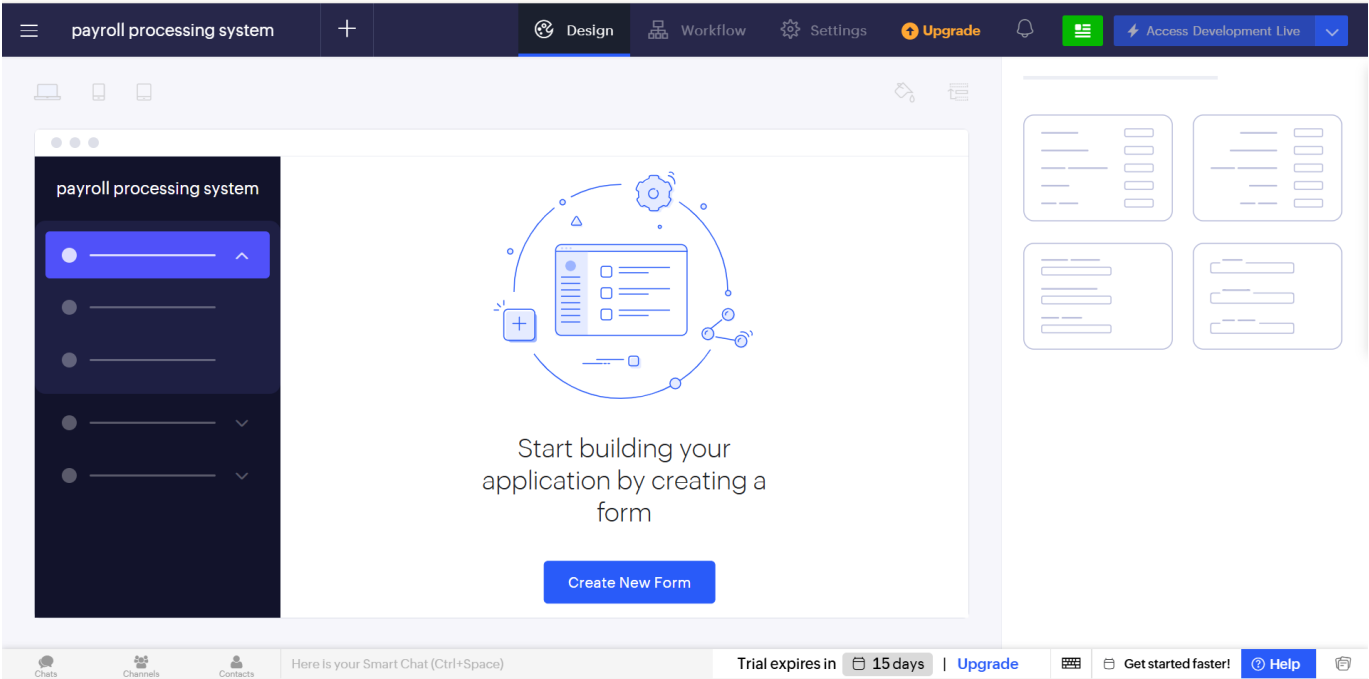
Application Name *

payroll processing system

☒ Enable Environment [Learn More](#)

Create Application

STEP 3: Create new application and create form from scratch



STEP 4: pay slip record Form

Enterprise Payroll Management System

Trial expires in 15 days Upgrade Edit this application Help

Enterprise Payroll Manage...

All Salary Structures

Dashboard

Records

Salary Structures

Salary Structure

All Salary Structures

Batches

Employee Masters

Employee	Basic Pay	HRA	DA	TA	Other Allowances	Deductions	Gross Pay
EMP001	\$ 52,000.00	\$ 13,000.00	\$ 4,600.00	\$ 3,100.00	\$ 2,100.00	\$ 5,200.00	\$ 67,500.00
EMP001 EMP002 EMP003 EMP004	\$ 60,000.00	\$ 15,000.00	\$ 5,400.00	\$ 3,600.00	\$ 2,400.00	\$ 6,000.00	\$ 76,800.00
EMP002 EMP003	\$ 48,000.00	\$ 12,000.00	\$ 4,200.00	\$ 2,800.00	\$ 1,800.00	\$ 4,800.00	\$ 62,000.00
EMP002 EMP003	\$ 55,000.00	\$ 13,750.00	\$ 5,000.00	\$ 3,300.00	\$ 2,200.00	\$ 5,500.00	\$ 70,450.00
EMP001 EMP002 EMP003 EMP004 EMP005	\$ 50,000.00	\$ 12,500.00	\$ 4,500.00	\$ 3,000.00	\$ 2,000.00	\$ 5,000.00	\$ 64,500.00

STEP 5: Creating transaction Form

Library Management System

Book Reservation

Field Properties

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Reservation Code

Student

Book

Reservation Date

Expected Pickup Date

Actual Pickup Date

Reservation Status

Notes

Field name

Reservation Code

Field link name

reservation_code

View Field References

Validation

Mandatory No duplicate values

Initial value

Initial value

Character maximum

255

Here is your Smart Chat (Ctrl+Space)

Trial expires in 15 days Upgrade

Get started faster! Help

STEP 6: Creating department reports

Enterprise Payroll Management S...
Departments

ORGANIZE RECORDS

Kanban Properties

Filter

Sorting

USER OPTIONS

Quick Filters

Custom Filters

Print & PDF

Kanban Properties

Choose how Information Is displayed In your Kanban report.

Group Columns By

Department

Empty Value Column

Show

Hide

Deleted Value Column

Show

Hide

Filter Columns By

None

Sort Columns By

None

Record Count Per Column

Show

Hide

Kanban report can display a maximum of 50 columns.

STEP 7: Creating payment status report

Enterprise Payroll Management S...
Payment Statuses

ORGANIZE RECORDS

Kanban Properties

Filter

Sorting

USER OPTIONS

Quick Filters

Custom Filters

Print & PDF

Kanban Properties

Choose how Information Is displayed In your Kanban report.

Group Columns By

Payment Status

Empty Value Column

Show

Hide

Deleted Value Column

Show

Hide

Filter Columns By

None

Sort Columns By

None

Record Count Per Column

Show

Hide

Kanban report can display a maximum of 50 columns.

1

Enterprise Payroll Management S...

Trial expires in 15 days

Upgrade

Get started faster

@Hale

STEP 8: Create Feedback Form

Enterprise Payroll Management System

Trial expires in 15 days

Upgrade

Edit this application

Help

Enterprise Payroll Manage...

All Salary Structures

Dashboard

Records

Salary Structures

Salary Structure

All Salary Structures

Batches

Employee Masters

Employee	Basic Pay	HRA	DA	TA	Other Allowances	Deductions	Gross Pay
EMP001	\$ 52,000.00	\$ 13,000.00	\$ 4,600.00	\$ 3,100.00	\$ 2,100.00	\$ 5,200.00	\$ 67,500.00
EMP001 EMP002 EMP003 EMP004	\$ 60,000.00	\$ 15,000.00	\$ 5,400.00	\$ 3,600.00	\$ 2,400.00	\$ 6,000.00	\$ 76,800.00
EMP002 EMP003	\$ 48,000.00	\$ 12,000.00	\$ 4,200.00	\$ 2,800.00	\$ 1,800.00	\$ 4,800.00	\$ 62,000.00
EMP002 EMP003	\$ 55,000.00	\$ 13,750.00	\$ 5,000.00	\$ 3,300.00	\$ 2,200.00	\$ 5,500.00	\$ 70,450.00
EMP001 EMP002 EMP003 EMP004 EMP005	\$ 50,000.00	\$ 12,500.00	\$ 4,500.00	\$ 3,000.00	\$ 2,000.00	\$ 5,000.00	\$ 64,500.00

STEP 10: Add sample data to all the created forms.

creatorapp.zoho.in/nandy.nandhinisri22/enterprise-payroll-management-system#Form:salary_structure

Enterprise Payroll Management System

Trial expires in 15 days

Upgrade

Edit this application

Enterprise Payroll Manage...

Salary Str

Employee Master

Employee Name

kani

First Name

Employee Code

6798

Department

IT

Designation

Intern

Date Of Joining

04-Aug-2026

Experience in Current Org

12

Overall Experience

2

Email Address

nandy.nandhinisri22@gmail.com

Employee

Basic Pay

HRA

DA

TA

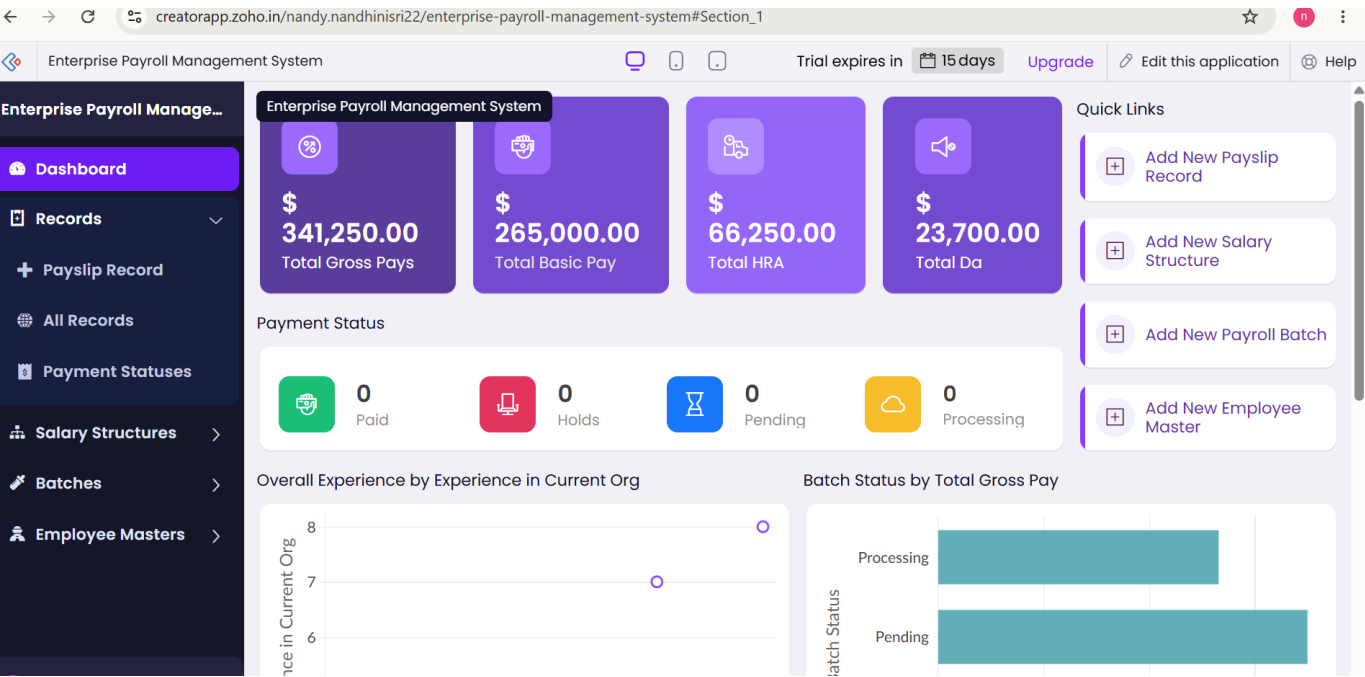
Other Allo

Deduction

Gross Pay

Net Pay

STEP 11: Create and display the Dashboard with important payslip record,payment status,salary structures and etc.



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

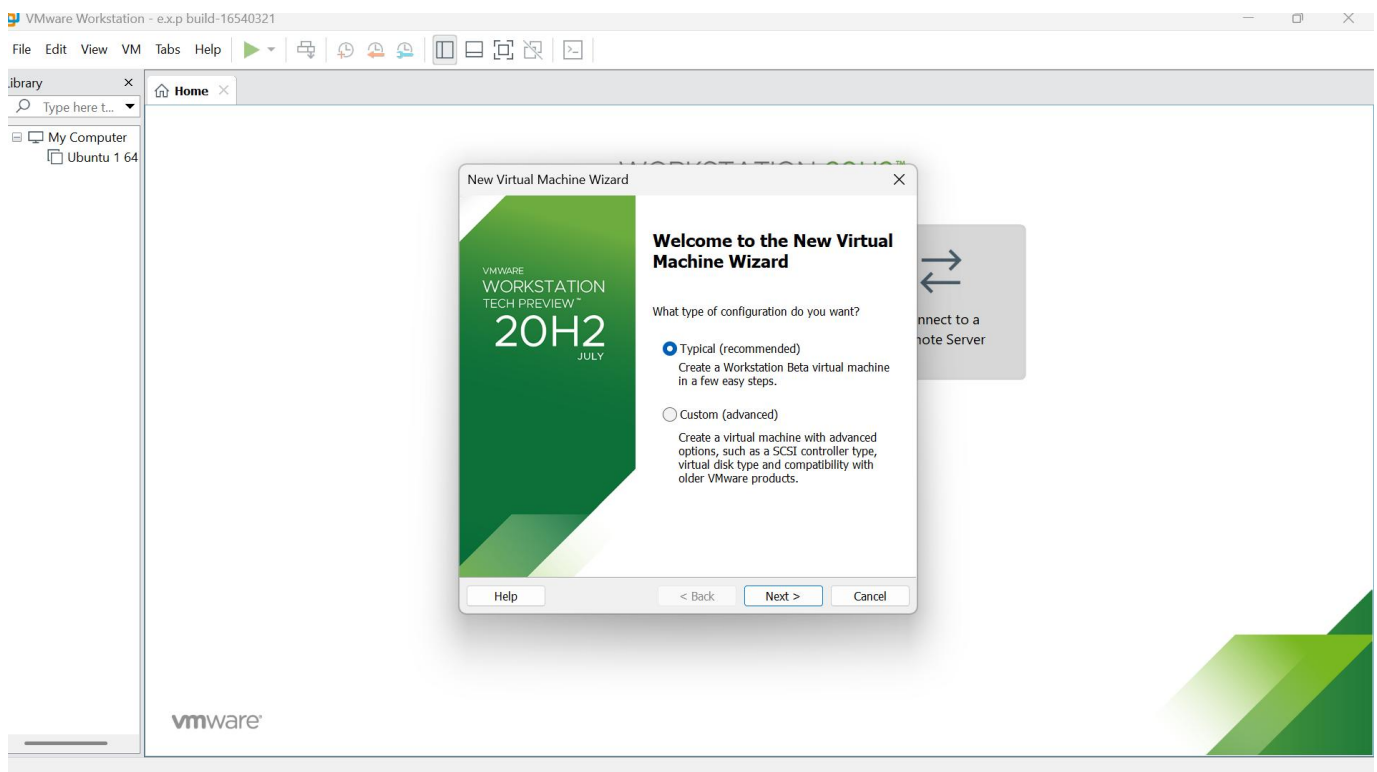
ANS:

AIM:

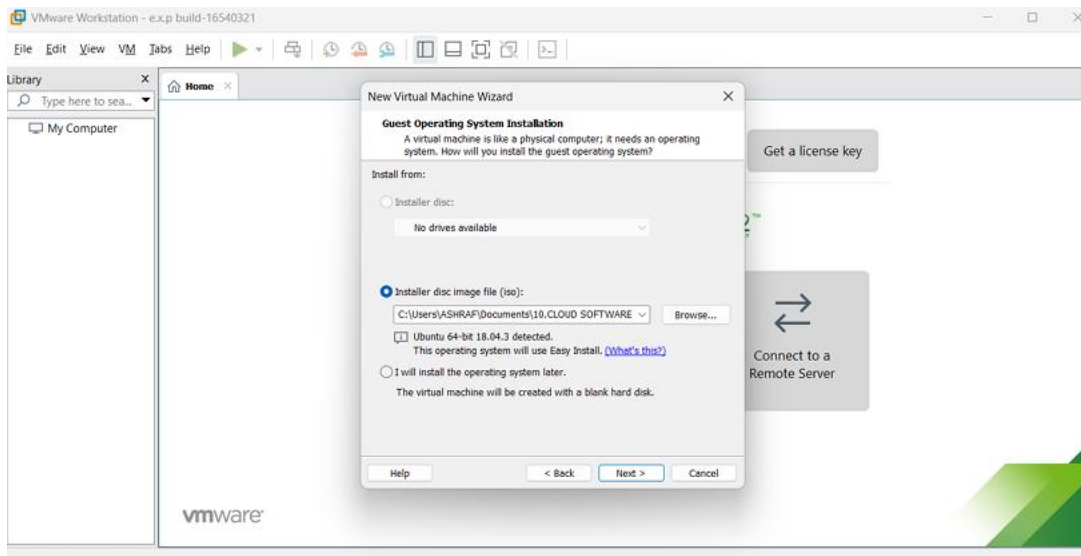
To demonstrate virtualization by installing a Type-2 Hypervisor (VMware Workstation) and creating and configuring a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage using a Host Operating System.

PROCEDURE:

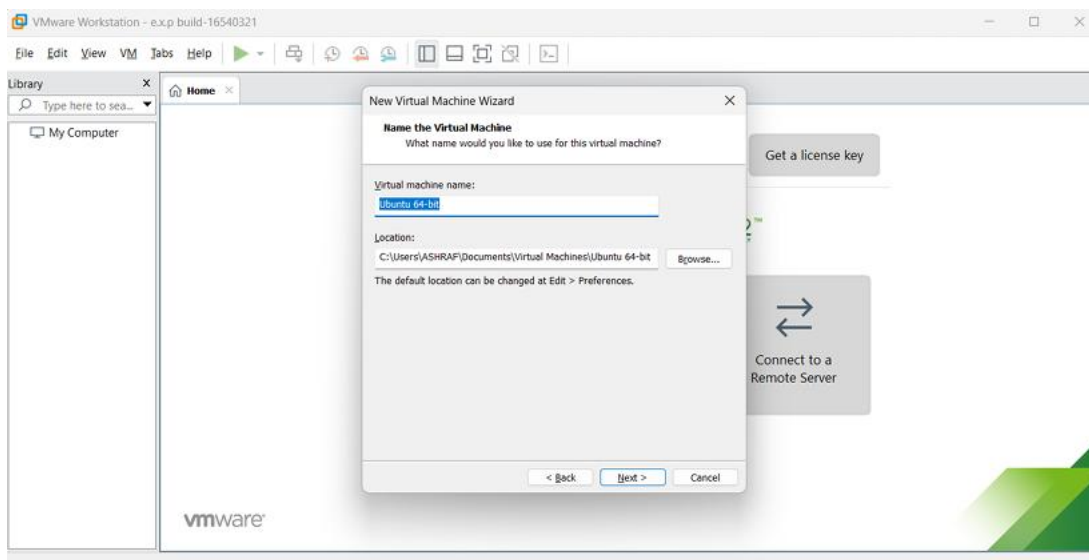
STEP 1:GOTO VM WARE WORKSTATION.



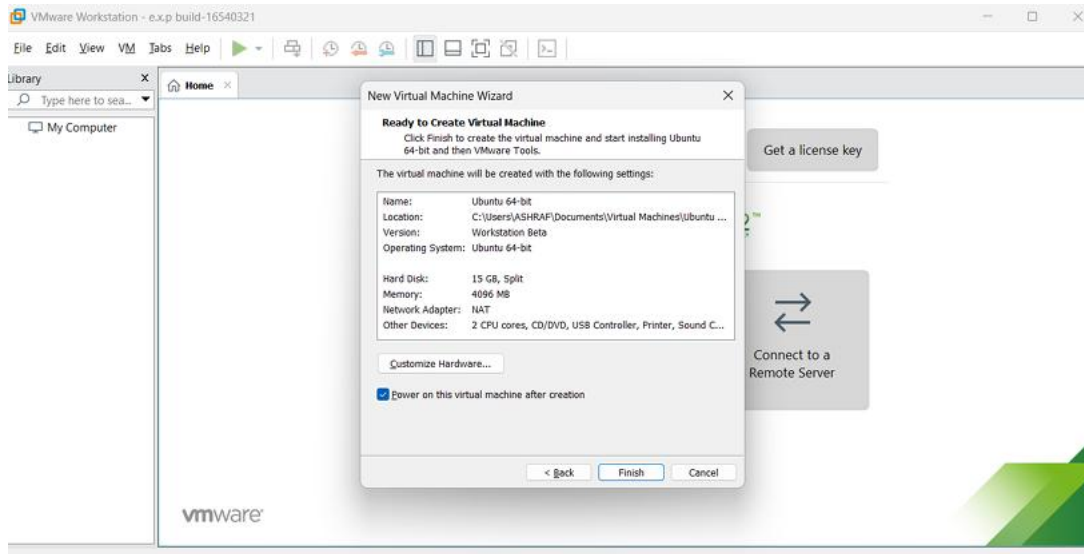
STEP 2: Select the ubuntu ISO files



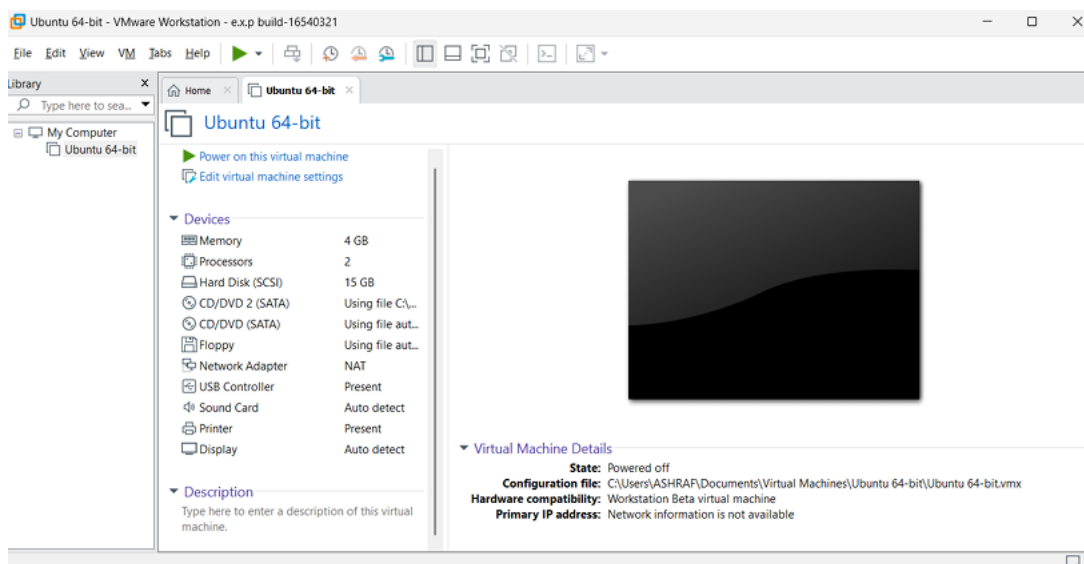
STEP 3: Set the VM name as Ubuntu-VM.



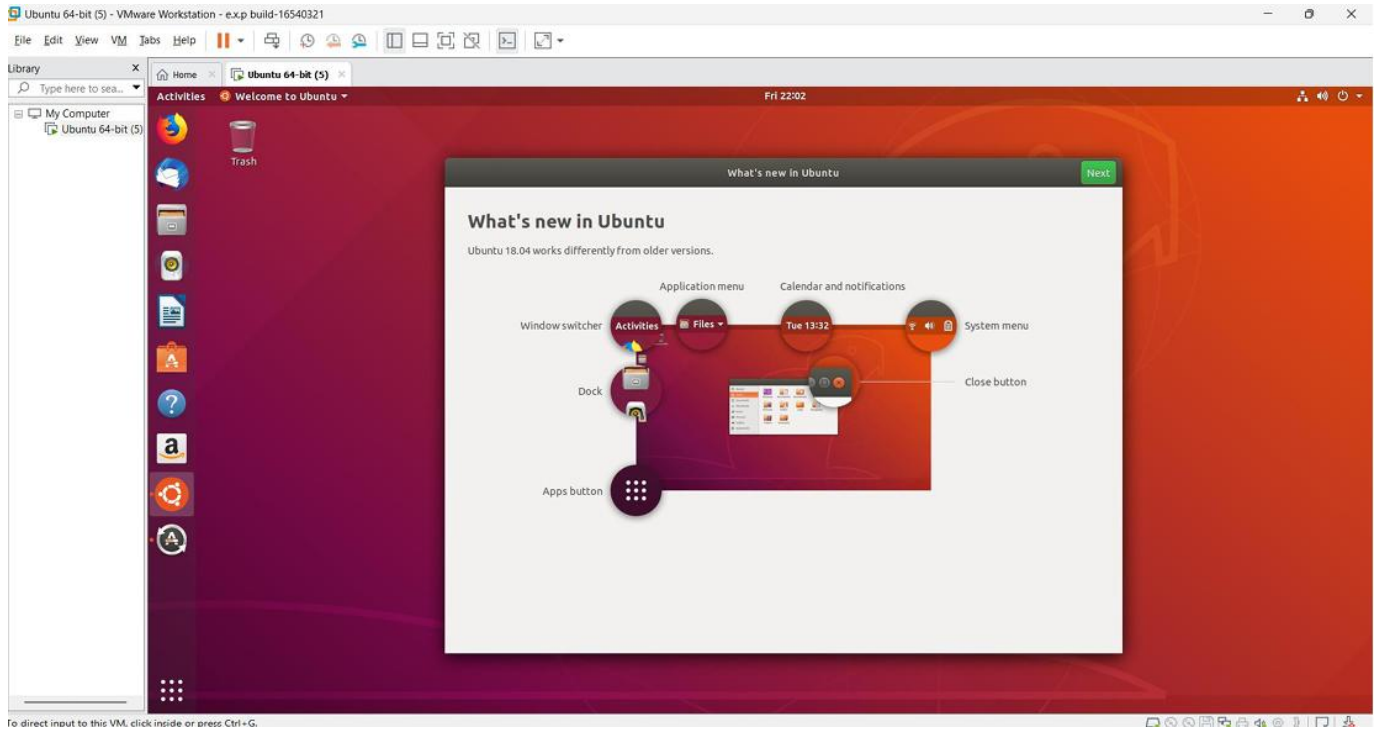
STEP 4: Set RAM = 2 GB and CPU = 1.



STEP 5: Set the hard disk size = 15 GB.



STEP 6: Start the VM and install Ubuntu



Result:

The virtual machine was successfully created using VMware with 1 CPU, 2 GB RAM and 15 GB storage.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

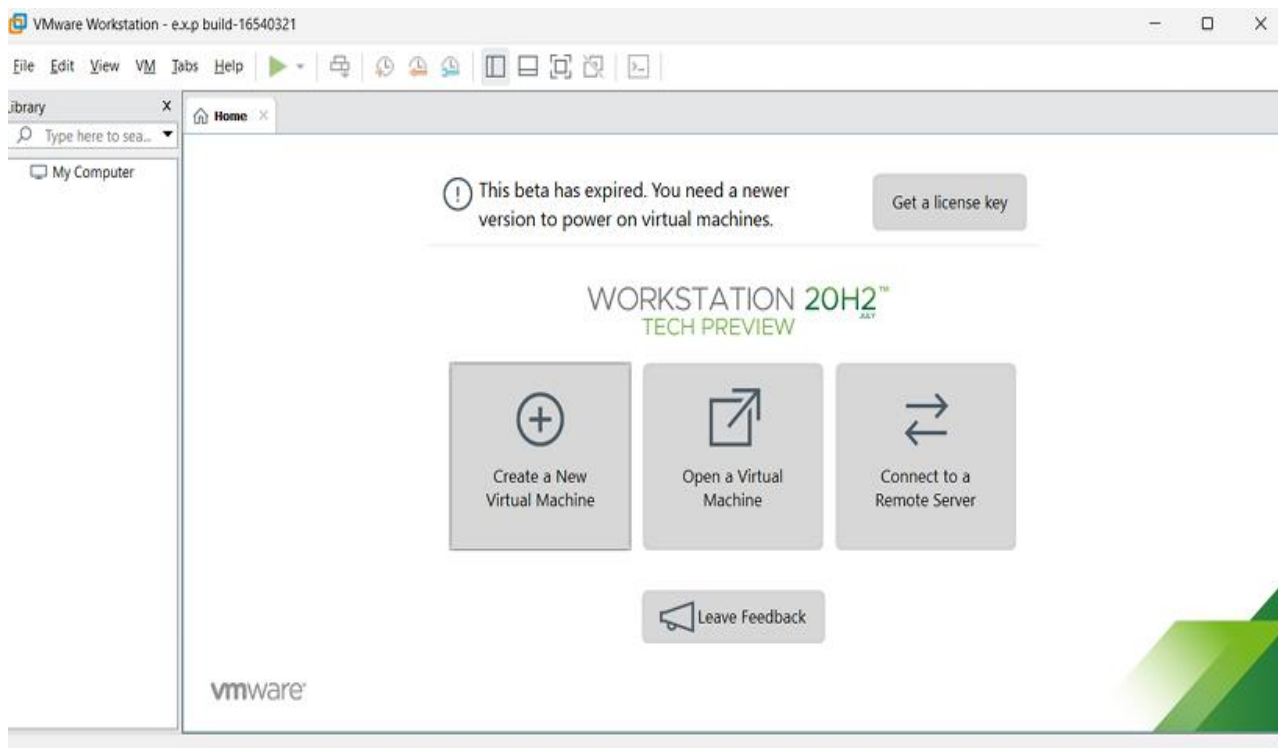
Ans:

AIM:

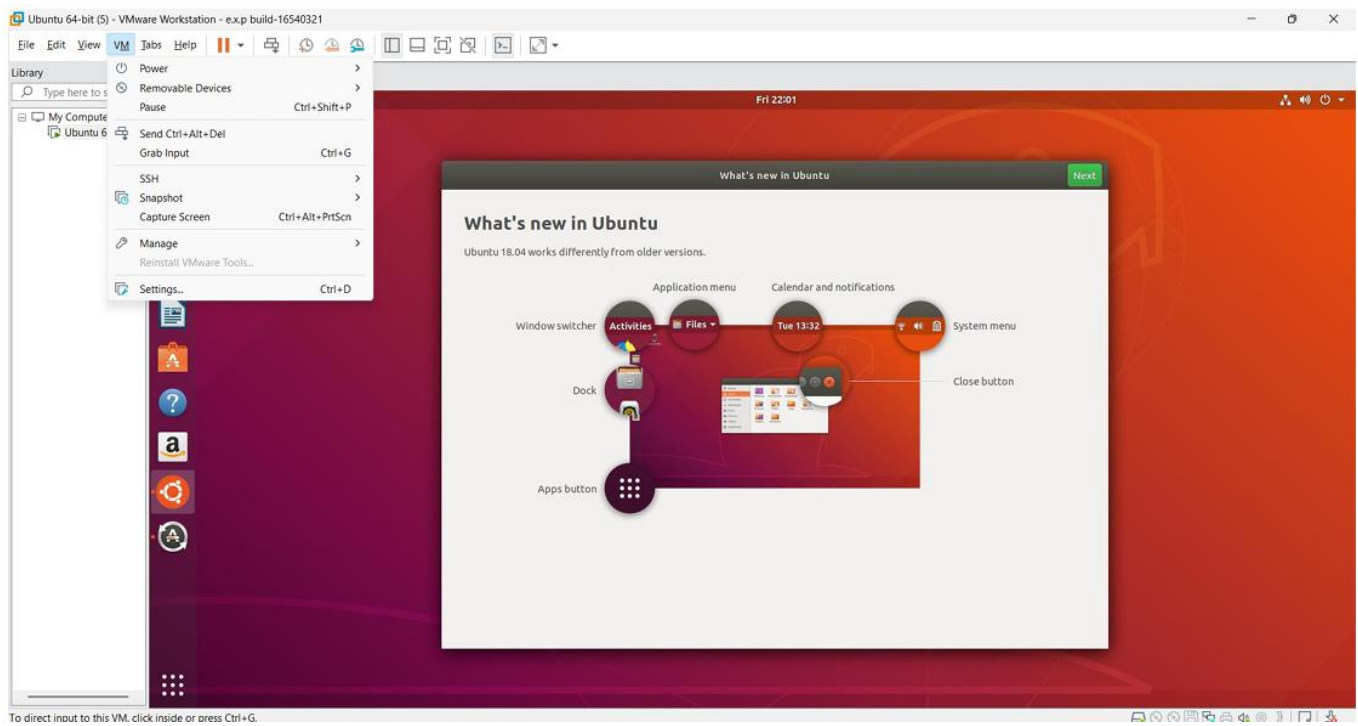
To create a clone of an existing virtual machine and test the cloned VM by restoring its previous version.

PROCEDURE:

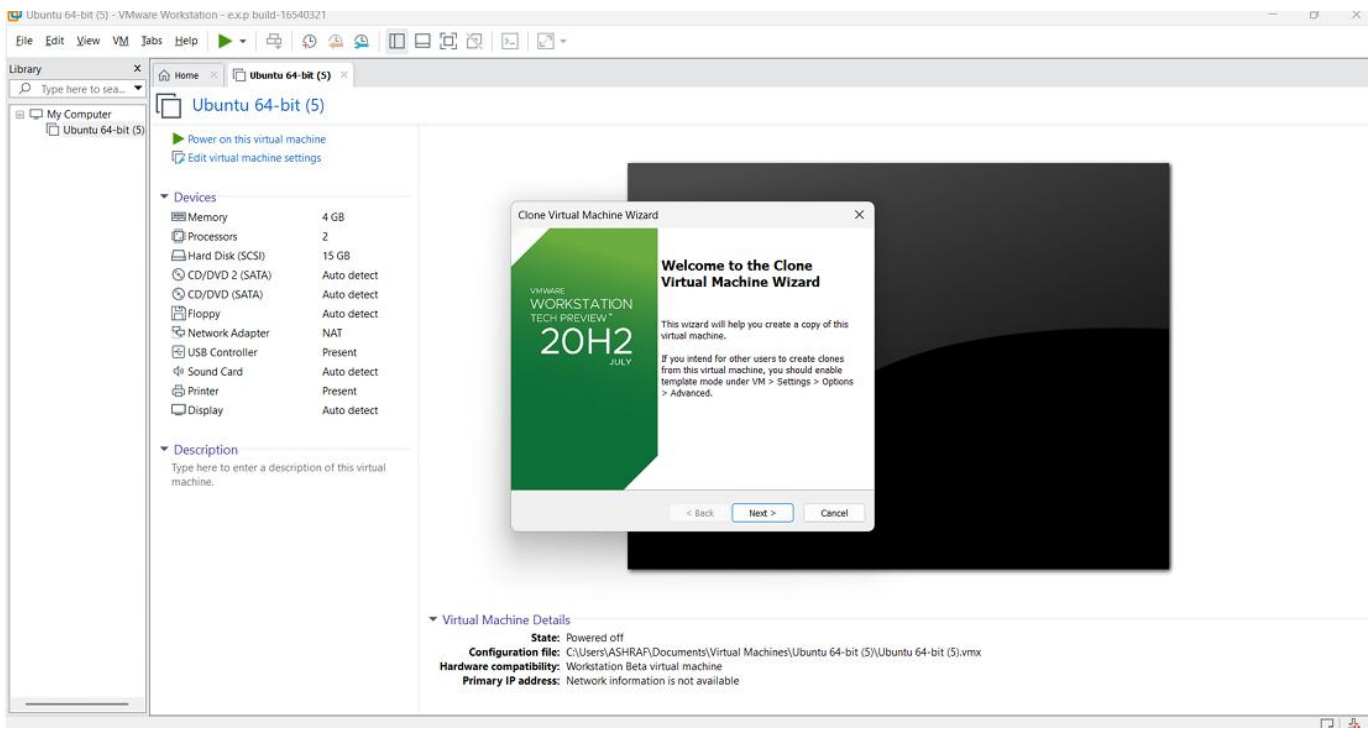
STEP 1: Open VMware Workstation, select the Ubuntu VM, and shut it down.



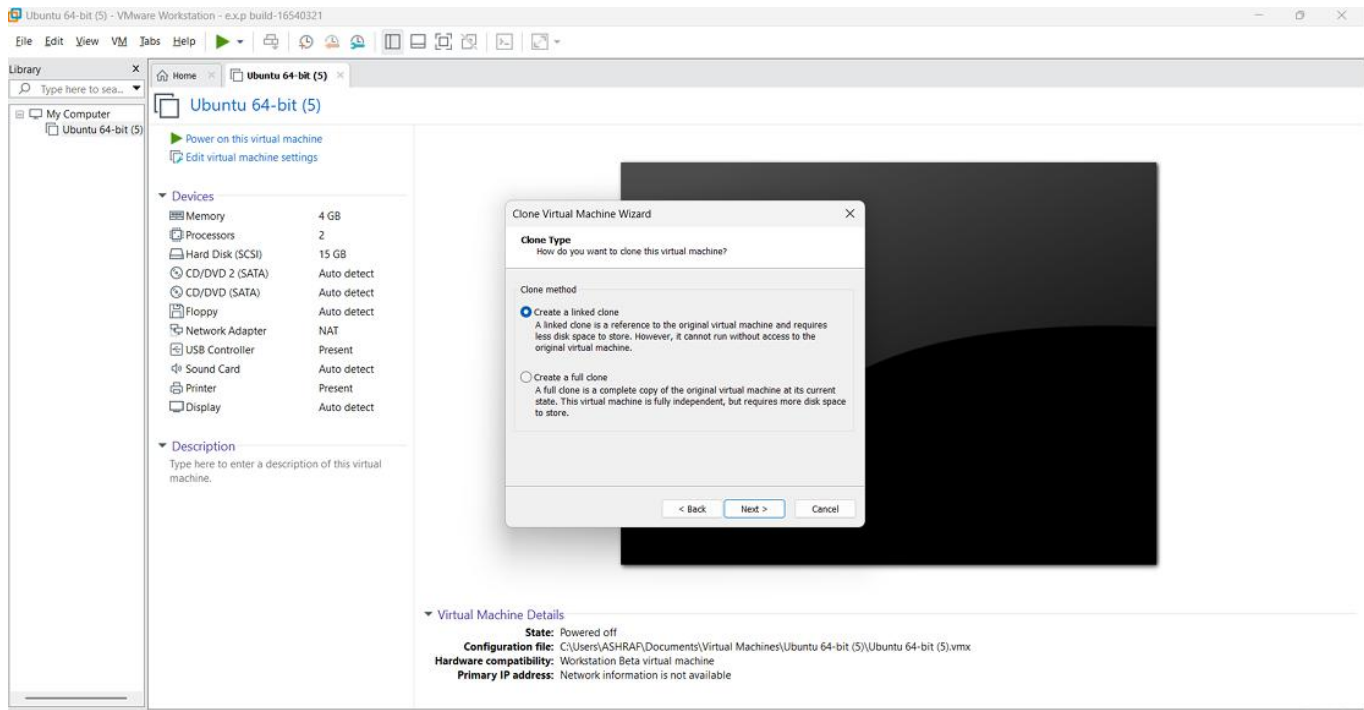
STEP 2: Select Manage → Clone to start the cloning process.



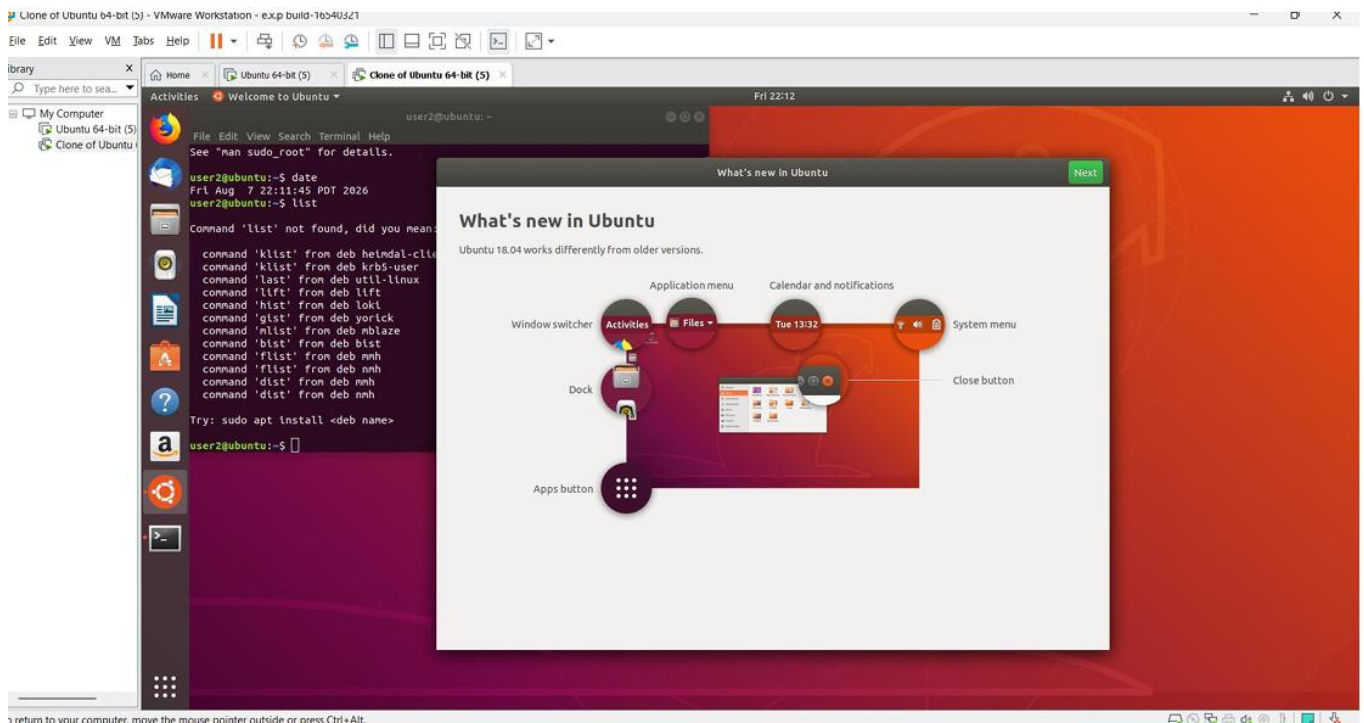
STEP 3: Select Full Clone and enter a name for the cloned VM



STEP 4: Complete the cloning process and start the cloned VM.



STEP 5: Restore the previous snapshot, start the VM, and verify that the previous state is loaded.



Result:

The VM was successfully cloned and tested, and its previous saved state was successfully restored

